

MC pileup profile -- does one profile serve signal and QCD?

CMS Simulation Preliminary

(13.6 TeV, 2024)

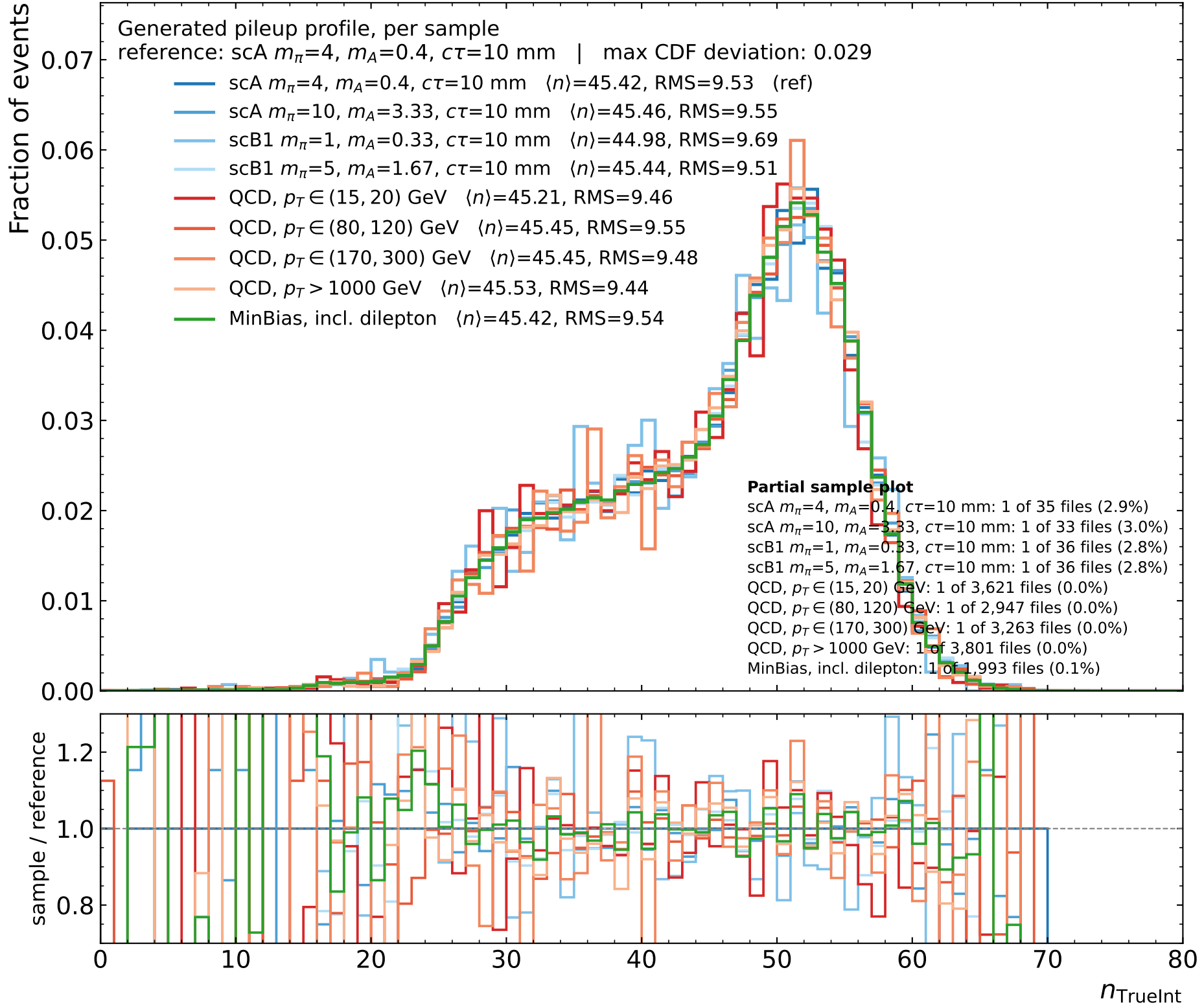
Generated pileup profile, per sample

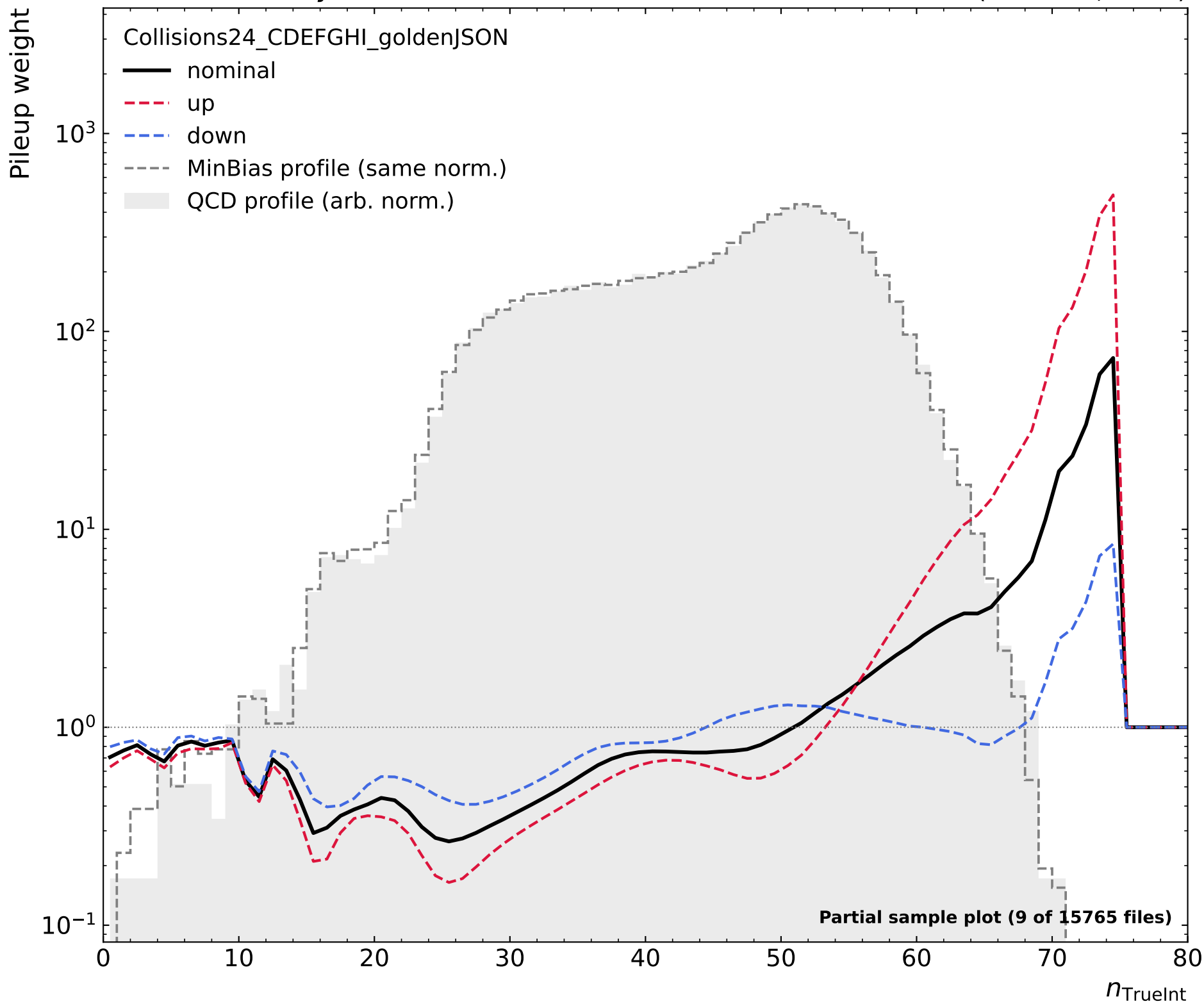
reference: scA $m_\pi=4$, $m_A=0.4$, $c\tau=10$ mm | max CDF deviation: 0.029

- scA $m_\pi=4$, $m_A=0.4$, $c\tau=10$ mm $\langle n \rangle=45.42$, RMS=9.53 (ref)
- scA $m_\pi=10$, $m_A=3.33$, $c\tau=10$ mm $\langle n \rangle=45.46$, RMS=9.55
- scB1 $m_\pi=1$, $m_A=0.33$, $c\tau=10$ mm $\langle n \rangle=44.98$, RMS=9.69
- scB1 $m_\pi=5$, $m_A=1.67$, $c\tau=10$ mm $\langle n \rangle=45.44$, RMS=9.51
- QCD, $p_T \in (15, 20)$ GeV $\langle n \rangle=45.21$, RMS=9.46
- QCD, $p_T \in (80, 120)$ GeV $\langle n \rangle=45.45$, RMS=9.55
- QCD, $p_T \in (170, 300)$ GeV $\langle n \rangle=45.45$, RMS=9.48
- QCD, $p_T > 1000$ GeV $\langle n \rangle=45.53$, RMS=9.44
- MinBias, incl. dilepton $\langle n \rangle=45.42$, RMS=9.54

Partial sample plot

- scA $m_\pi=4$, $m_A=0.4$, $c\tau=10$ mm: 1 of 35 files (2.9%)
- scA $m_\pi=10$, $m_A=3.33$, $c\tau=10$ mm: 1 of 33 files (3.0%)
- scB1 $m_\pi=1$, $m_A=0.33$, $c\tau=10$ mm: 1 of 36 files (2.8%)
- scB1 $m_\pi=5$, $m_A=1.67$, $c\tau=10$ mm: 1 of 36 files (2.8%)
- QCD, $p_T \in (15, 20)$ GeV: 1 of 3,621 files (0.0%)
- QCD, $p_T \in (80, 120)$ GeV: 1 of 2,947 files (0.0%)
- QCD, $p_T \in (170, 300)$ GeV: 1 of 3,263 files (0.0%)
- QCD, $p_T > 1000$ GeV: 1 of 3,801 files (0.0%)
- MinBias, incl. dilepton: 1 of 1,993 files (0.1%)

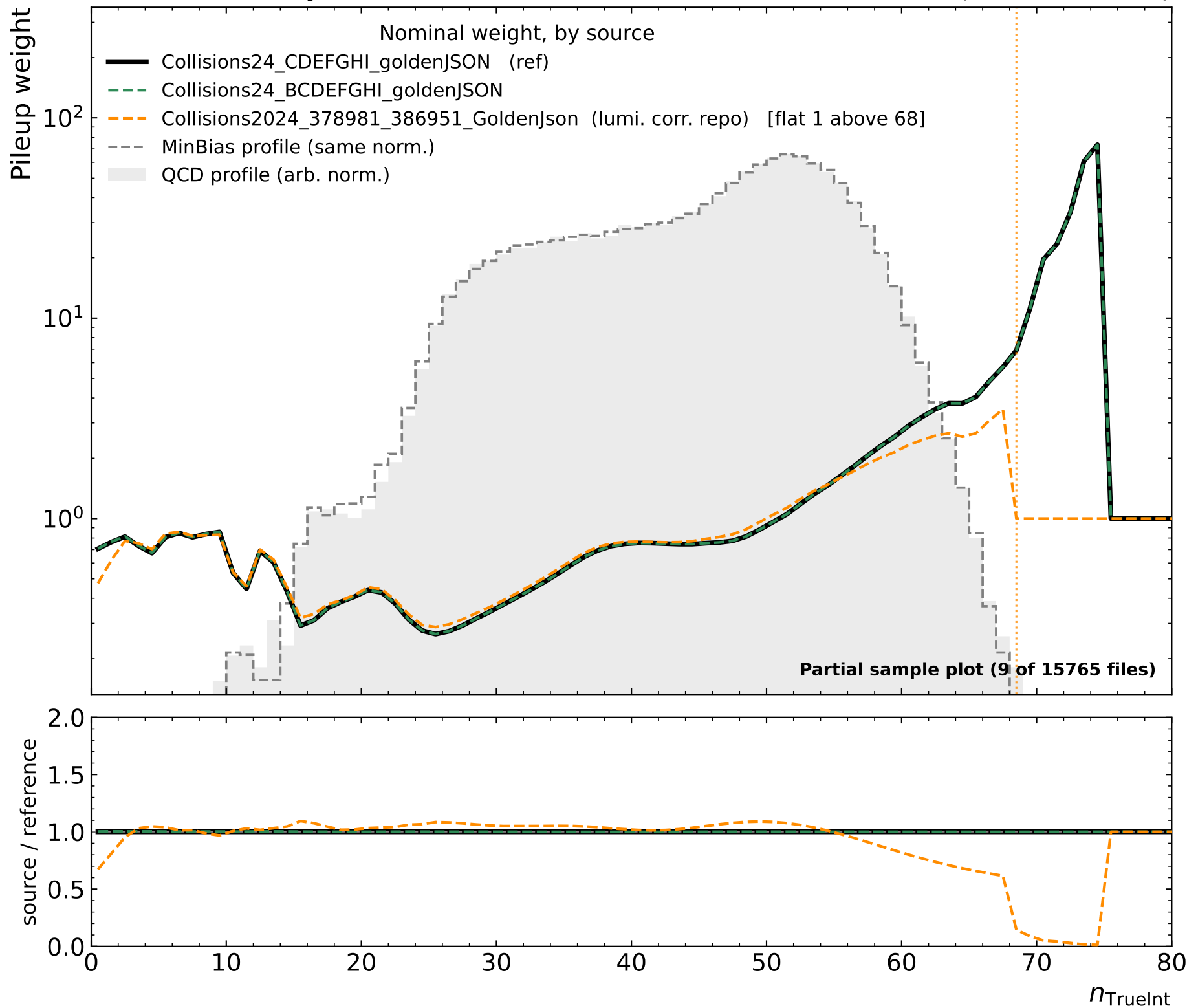




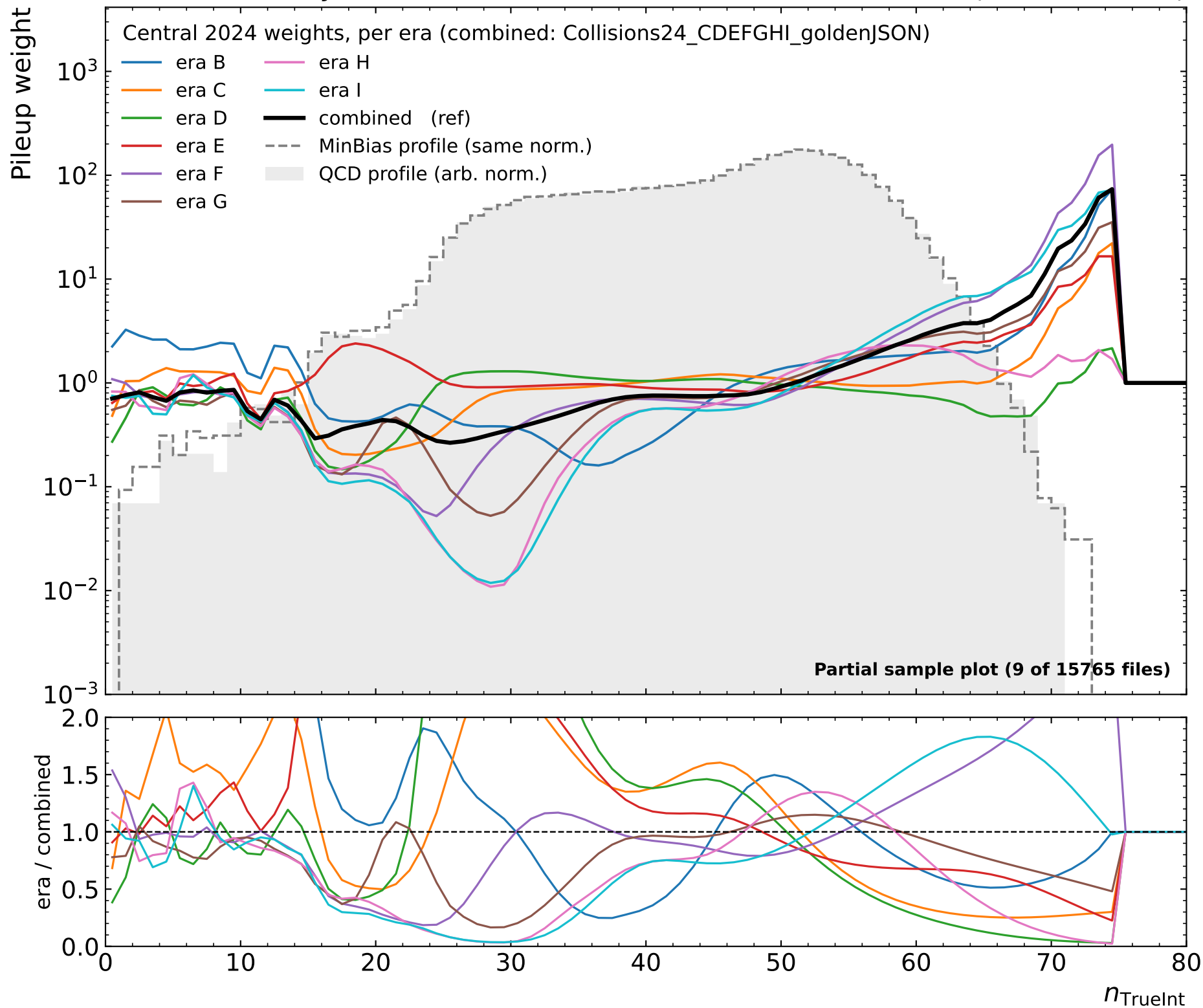
Weight sources compared -- central vs the lum-corrections stopgap

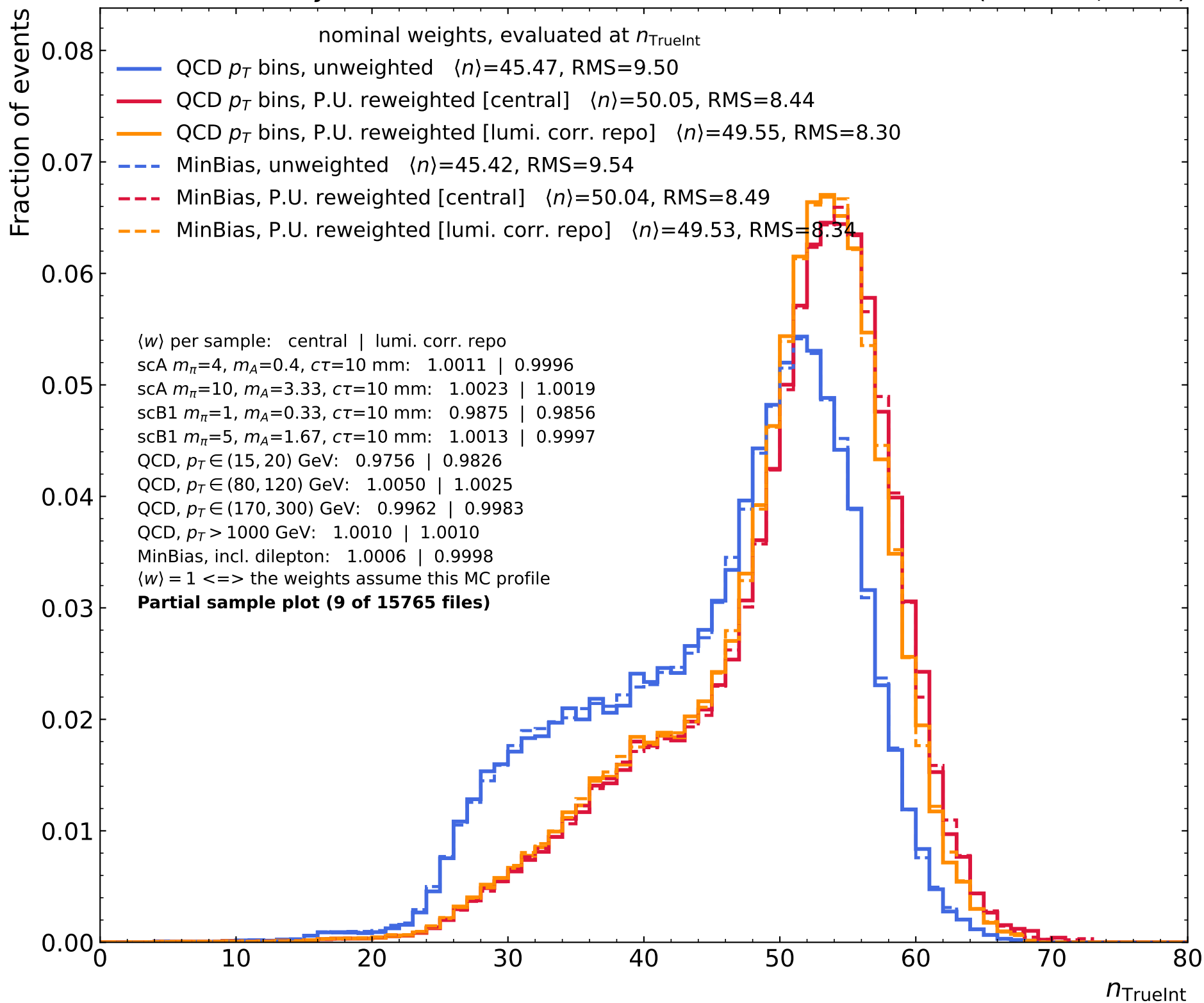
CMS Preliminary

(13.6 TeV, 2024)

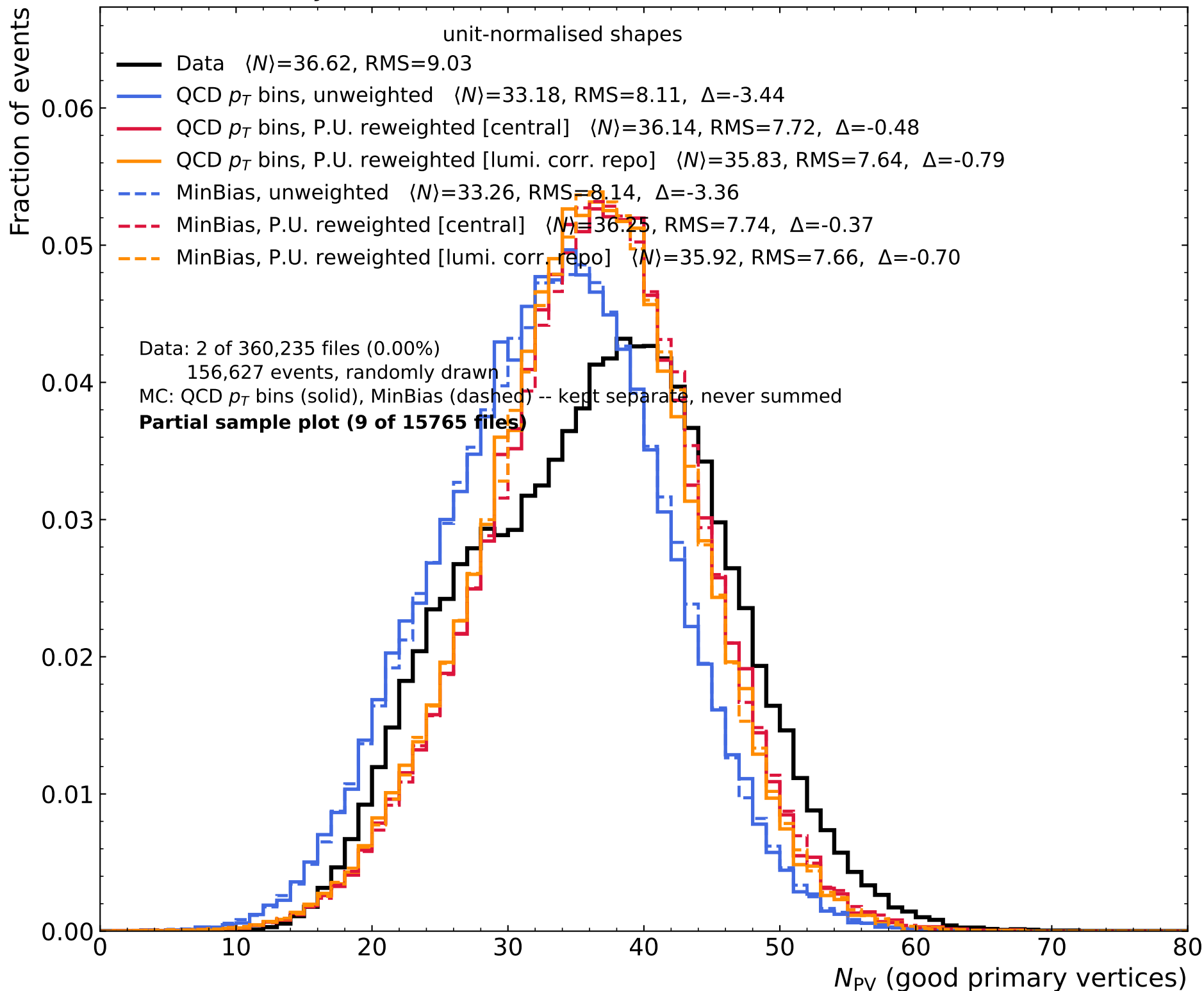


Per-era pileup weights -- the handle a single merged histogram cannot give
CMS Preliminary (13.6 TeV, 2024)





Data validation -- does the pileup correction improve the agreement?

CMS Preliminary0.00061 fb⁻¹ (13.6 TeV, 2024)

Data pileup by era -- is the merged data/MC offset an averaging effect?

CMS Preliminary

0.00061 fb⁻¹ (13.6 TeV, 2024)

